

Summary

Autoimmune diseases arise when the immune system loses tolerance to self-components and begins attacking the body's own organs, tissues or cells, causing damage and a variety of clinical syndromes. It is estimated that 7.6–9.4 per cent of the global population have one or more autoimmune conditions. These diseases are rarely curable; most patients require long-term or lifelong medication. Certain conditions, such as lupus nephritis, can be life-threatening and severely diminish quality of life.

Historically, first-line therapy comprised glucocorticoids and non-steroidal anti-inflammatory drugs, but both carry significant side-effects and relapse is common. As understanding of cytokine-mediated mechanisms in inflammation and autoimmunity has advanced, biologic agents targeting specific cytokines have been developed, inaugurating a new market segment. Because many autoimmune diseases share pathogenic pathways, these biologics often prove effective across multiple indications.

Between 2009 and 2019, therapeutics targeting certain immune responses in the global Top 100 drug list rose from 8 per cent to 27 per cent. In 2019, 25 of these drugs featured among the world's top 100 sellers, and the market is projected to reach CNY 1330 billion by 2030. Improved diagnostic technologies and deeper disease understanding are enabling more patients to be correctly diagnosed, further expanding market potential.

In China, the market for autoimmune disease therapeutics grew from CNY 90 billion in 2014 to CNY 134 billion in 2018, with forecasts pointing to CNY 1330 billion by 2030. Biologics have driven most of this growth, posting a compound annual growth rate of 21.1 per cent, compared with just 8.6 per cent for small-molecule drugs. Broader health-insurance coverage and greater patient awareness bode well for the future of biologic therapies.

Key Takeaway

- **Impact of Reimbursement on Biologics**

Inclusion in health-insurance schemes exerts a decisive influence on the domestic market for biologic agents. Prior to reimbursement, adalimumab—the world's top-selling drug for several years—saw declining sales in China; within six months of

being covered, its half-year domestic revenue quadrupled the total for the previous full year. Domestic biosimilars benefit from substantial price advantages and widespread insurance inclusion, and are already beginning to replace originator biologics whose patents have expired.

- **Emergence of New Cytokine Targets**

Beyond TNF, interleukin-6, IL-17 and IL-23 have become focal points for therapeutics, yielding blockbuster agents such as Ustekinumab. Several head-to-head trials are underway comparing these new drugs with adalimumab and other interleukin-targeting therapies to evaluate relative efficacy. Meanwhile, JAK inhibitors have garnered intense interest, and multiple oral formulations have reached the market. Developing drugs tailored to the advantages of specific indications has become a clear trend.

- **Rapid Expansion of Indications**

Since cytokine targets often play crucial roles across different diseases, many companies continue to extend their drugs' approved uses post-launch. For example, adalimumab's indications have grown from rheumatoid arthritis to encompass most common autoimmune conditions. Even in the face of strong competition across each indication, this strategy has driven continued sales growth. The speed at which new indications are added has thus become a key determinant of a drug's market success.

- **Safety and Clinical Nuance**

Autoimmune diseases typically require chronic therapy, and interventions against TNF, interleukins and JAK pathways involve broad immunomodulation. Safety is therefore under constant scrutiny. Given the extensive crosstalk among cytokines, different inhibition strategies—partial versus complete blockade, ligand versus receptor targeting—may produce unexpected clinical outcomes. Thorough evaluation of clinical data is essential to discern true benefit–risk profiles.

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